

Relevance-Gated Empirical Study

Owner-Side Analysis, Computation, and Results Summary - V3.5

Prepared bundle contents: reproducible code, formula/evaluation scripts, joined audit files, diagnostics, AUC calculations, and current-best empirical results.

1. Interpretation frame

The framework outputs are not intended to represent an all-knowing oracle. In particular, $T_{abs}(s)$ is interpreted as the evidence-weighted posterior that a highly unbiased, fairly skeptical expert would assign using the evidence available to the scorer. This is different from metaphysical certainty or omniscient truth.

The current-best analysis uses V3.2 formula-corrected HumAID/CredEvent results and the V3.5 LIAR extensive-research scorer output.

2. Dataset-to-output mapping

Dataset	Linked output	What it tests
HumAID	$I(s,r)$	Receiver-dependent importance for a crisis-information monitor. It stresses relevance gating, broad humanitarian stakes, role-based consequence, and operational crisis importance.
CredEvent	$T_{perc}(s,r)$	Perceived truth/credibility of event claims by screened crowd raters. It stresses perceived credibility, delivery context, event-stream corroboration, and receiver interpretation.
LIAR / PolitiFact	$T_{abs}(s)$	Evidence-weighted truth probability. It stresses source credibility, claim specificity/falsifiability, consistency with established knowledge, independent corroboration, and direct evidence.

3. Current-best main metrics

Dataset	Output	n	Spearman	Pearson	MAE	Graded AUC / C-index	Primary binary AUC	Binary threshold	Model mean	Target mean
HumAID	$I(s,r)$	200	0.407	0.477	0.158	0.658	0.757	0.500	0.813	0.785
CredEvent_TurkRatings	$T_{perc}(s,r)$	200	0.591	0.563	0.124	0.719	0.876	0.900	0.882	0.938
LIAR_PolitiFact	$T_{abs}(s)$	200	0.630	0.560	0.263	0.764	0.825	0.500	0.436	0.390

Reading the metrics: Spearman and graded AUC describe ranking behavior; Pearson describes linear association; MAE and mean comparison describe calibration. The results are strongest for CredEvent and LIAR ranking behavior, while HumAID is better calibrated in mean but still has false-positive issues for disaster-adjacent non-operational posts.

4. AUC diagnostics

Dataset	Output	Threshold	Value	AUC	Positives	Negatives
HumAID	$I(s,r)$	target ≥ 0.5	0.500	0.757	180	20
HumAID	$I(s,r)$	target ≥ 0.75	0.750	0.773	140	60
HumAID	$I(s,r)$	target ≥ 0.9	0.900	0.696	120	80

CredEvent_TurkRatings	T_perc(s,r)	target >= 0.5	0.500	undefined	200	0
CredEvent_TurkRatings	T_perc(s,r)	target >= 0.9	0.900	0.876	169	31
CredEvent_TurkRatings	T_perc(s,r)	median	0.953	0.791	108	92
LIAR_PolitiFact	T_abs(s)	target >= 0.25	0.250	0.823	132	68
LIAR_PolitiFact	T_abs(s)	target >= 0.5	0.500	0.825	64	136
LIAR_PolitiFact	T_abs(s)	target >= 0.75	0.750	0.801	30	170

AUC is threshold-dependent because the targets are graded rather than purely binary. The graded AUC/C-index avoids a single threshold by measuring pairwise ranking concordance over continuous or ordinal targets.

5. Output descriptions

T_abs(s): Evidence-weighted truth-probability. It uses source credibility, statement properties, corroboration, established knowledge, and empirical evidence. It is receiver-independent, but it is still evidence-limited: it models a skeptical expert using available evidence, not an all-knowing model.

T_perc(s,r): Perceived truth-probability for a receiver. It starts from T_abs and updates through delivery context, relevance-gated situational context, and receiver properties. CredEvent is the current empirical proxy for this output.

I(s,r): Receiver-dependent importance. It asks how consequential or attention-worthy a statement is for a specified receiver. A false statement can still be important if it creates high stakes, urgency, harm, or decision relevance. HumAID is the current empirical proxy for this output.

6. Dataset-specific interpretation

6.1 HumAID - I(s,r)

HumAID tests whether the model identifies humanitarian/crisis-monitoring importance. After the V3.2 formula correction, the model mean is close to the target mean, but ranking is moderate. The remaining failure mode is false-positive inflation: posts that mention disasters or public events but do not carry real humanitarian or operational relevance can still receive excessive importance.

HumAID group means

hidden_class_label	n	target_mean	model_mean	mae
requests_or_urgent_needs	20	1.000	0.869	0.131
injured_or_dead_people	20	1.000	0.912	0.088
missing_or_found_people	20	0.950	0.915	0.043
displaced_people_and_evacuations	20	0.950	0.865	0.100
caution_and_advice	20	0.900	0.892	0.082
infrastructure_and_utility_damage	20	0.900	0.780	0.140
rescue_volunteering_or_donation_effort	20	0.800	0.915	0.115
other_relevant_information	20	0.700	0.751	0.235
sympathy_and_support	20	0.500	0.590	0.150
not_humanitarian	20	0.150	0.637	0.494

6.2 CredEvent - T_perc(s,r)

CredEvent is the cleanest result. It tests perceived credibility/truth for social-media event claims, and the V3 scoring/calibration instructions substantially fixed the earlier underuse of public event-stream delivery and corroboration cues. A standard target >= 0.50 ROC-AUC is undefined because all targets are above 0.50; the useful binary task is separating very credible from less credible claims, where AUC is strong.

6.3 LIAR / PolitiFact - T_abs(s)

LIAR tests evidence-weighted truth, so performance depends heavily on real row-specific evidence retrieval. Broad context alone failed. Outside research improved the result, and V3.5 extensive research plus explicit anti-overconservatism instructions produced the best LIAR run. It ranks labels in the correct order but still compresses the tails: very false claims are too high and fully true claims remain too low.

LIAR label-group diagnostics

hidden_label	n	target_mean	model_mean	mae
pants-fire	34	0.005	0.135	0.134
false	34	0.100	0.196	0.225
barely-true	34	0.250	0.344	0.351
half-true	34	0.400	0.510	0.364
mostly-true	34	0.700	0.697	0.293
true	30	0.950	0.776	0.204

7. Computation/code contents in the archive

The bundle includes `formula_engine_relevance_gated_v3_2.py` and `evaluate_external_outputs_v3_2.py`, extracted from the V3.2 owner patch package. It also includes `recompute_metrics_auc_and_diagnostics_v3_5.py`, which recomputes the current-best metrics, binary AUCs, and graded AUC/C-index from the joined audit files.

The archive does not include hidden private reasoning or chain-of-thought. It includes the reproducible owner-side scripts and result files available from the working directory, plus a new metrics/AUC reproduction script for transparency.

8. Current overall conclusion

The current empirical story is: CredEvent provides the strongest and cleanest support for T_perc(s,r); HumAID provides useful but imperfect support for I(s,r), with good mean calibration but moderate ranking; LIAR provides a meaningful T_abs(s) diagnostic after extensive research, with strong ranking improvement but remaining calibration compression at the truth-value extremes.

9. Important leakage warning

This is an owner-side analysis package. It contains target-joined audit files and diagnostics. It should not be sent to external scorers. Scorers should receive only the no-target scorer packet archives.

End of summary - V3.5